

ASSESSMENT TOOL 3: Dataset Integration

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO3: Identify and clean inconsistencies in data by handling missing values, duplicates, outliers, formatting issues, and applying techniques like fuzzy matching and regex.

Assessment Tool Weightage: 30%

Total Marks: 20

Code of Conduct

I Naresh M (192424381) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Naresh M

Dataset Integration Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Dataset Understanding & Preparation	30%	Correctly understands both datasets, identifies relevant columns/keys, and loads them properly.	Minor misunderstanding or preparation issue.	Loads datasets but needs some guidance in identifying relevant fields.	Unable to identify or prepare the datasets correctly.
Dataset Integration	20%	Correctly integrates datasets using appropriate merge(), join(), or concat() and selects the correct integration key.	Integration is mostly correct with minor errors.	Performs integration with significant guidance or minor key/operation errors.	Unable to integrate the datasets correctly.
Data Processing & Analysis	15%	Correctly performs all required calculations, filtering, grouping, or transformations on the integrated data.	Performs most required analysis correctly.	Performs basic analysis but misses some required operations.	Analysis is incorrect or incomplete.

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Python/Pandas Implementation	20%	Code is correct, efficient, well-structured, and uses appropriate Pandas functions.	Code works with minor inefficiencies/errors.	Code works partially and requires corrections.	Code contains major errors or does not execute.
Output & Interpretation	15%	Clearly presents the required output and correctly interprets the results.	Output is correct with limited interpretation.	Output is partially correct or poorly presented.	Output is missing/incorrect and results are not interpreted.

Total Marks Awarded:

Signature of the Course Faculty

Questions:

Set 2 – Dataset Integration Tasks

1. Customer–Order Integration

You are given:

- customer.csv — Customer_ID, Customer_Name, City
- orders.csv — Order_ID, Customer_ID, Order_Date, Order_Amount

Task: Merge the datasets using Customer_ID. Display customer name, city, order date, and order amount. Find the customer with the highest total order amount.

```
Customer_ID,Customer_Name,City
C101,Raj Kumar,Chennai
C102,Priya S,Coimbatore
C103,Arun V,Salem
C104,Kavya R,Madurai
```

```
Order_ID,Customer_ID,Order_Date,Order_Amount
0001,C101,2025-01-10,15000
0002,C102,2025-01-11,12000
0003,C103,2025-01-12,8000
0004,C101,2025-01-15,20000
0005,C104,2025-01-18,5000
```

```
CO3 AT2 1.py > ...
1  import pandas as pd
2
3  customers = pd.read_csv("customer.csv")
4  orders = pd.read_csv("orders.csv")
5
6  merged = pd.merge(orders, customers, on="Customer_ID")
7  print("Merged Data:\n", merged[['Customer_Name', 'City', 'Order_Date', 'Order_Amount']])
8
9  total_orders = merged.groupby("Customer_Name")['Order_Amount'].sum()
10 highest_customer = total_orders.idxmax()
11 print("\nCustomer with Highest Total Order Amount:", highest_customer, "-", total_orders.max())
```

Output:

```
Merged Data:
   Customer_Name  City  Order_Date  Order_Amount
0    Raj Kumar   Chennai  2025-01-10         15000
1    Priya S  Coimbatore  2025-01-11         12000
2    Arun V    Salem  2025-01-12          8000
3    Raj Kumar   Chennai  2025-01-15         20000
4    Kavya R    Madurai  2025-01-18          5000

Customer with Highest Total Order Amount: Raj Kumar - 35000
```

2. Employee–Attendance Integration

You are given:

- employee.csv — Employee_ID, Employee_Name, Department
- attendance.csv — Employee_ID, Date, Status

Task: Integrate the two datasets using Employee_ID. Calculate the number of days each employee was present and absent.

```
Employee_ID,Date,Status
E101,2025-01-10,Present
E101,2025-01-11,Absent
E102,2025-01-10,Present
E102,2025-01-11,Present
E103,2025-01-10,Absent
E103,2025-01-11,Present
E104,2025-01-10,Present
E104,2025-01-11,Absent
```

```
Employee_ID,Employee_Name,Department
E101,Ravi Kumar,HR
E102,Priya S,Finance
E103,Arun V,IT
E104,Kavya R,Marketing
```

```
CO3 AT2 2.py > ...
1  import pandas as pd
2  employees = pd.read_csv("employee.csv")
3  attendance = pd.read_csv("attendance.csv")
4  merged = pd.merge(attendance, employees, on="Employee_ID")
5  summary = merged.groupby("Employee_Name")["Status"].value_counts().unstack().fillna(0)
6  print("Merged Data:\n", merged[['Employee_Name', 'Department', 'Date', 'Status']])
7  print("\nAttendance Summary:\n", summary)
```

Output:

```

Merged Data:
  Employee_Name Department      Date  Status
0   Ravi Kumar      HR  2025-01-10  Present
1   Ravi Kumar      HR  2025-01-11  Absent
2    Priya S      Finance  2025-01-10  Present
3    Priya S      Finance  2025-01-11  Present
4    Arun V        IT  2025-01-10  Absent
5    Arun V        IT  2025-01-11  Present
6   Kavya R  Marketing  2025-01-10  Present
7   Kavya R  Marketing  2025-01-11  Absent

Attendance Summary:
  Status      Absent  Present
Employee_Name
Arun V          1.0      1.0
Kavya R          1.0      1.0
Priya S          0.0      2.0
Ravi Kumar       1.0      1.0

```

3. Course-Student Integration

You are given:

- students.csv — Student_ID, Student_Name, Department
- enrollment.csv — Student_ID, Course_ID, Course_Name, Credits

Task: Merge the datasets using Student_ID. Display student name, department, course name, and credits. Find the total number of credits registered by each student.

```

Student_ID,Student_Name,Department
S101,Raj Kumar,CS
S102,Priya S,IT
S103,Arun V,EEE
S104,Kavya R,MECH

```

```

Student_ID,Course_ID,Course_Name,Credits
S101,C001,Data Structures,4
S101,C002,Algorithms,3
S102,C003,Database Systems,4
S103,C004,Circuits,3
S104,C005,Thermodynamics,4
S104,C006,Fluid Mechanics,3

```

```

CO3 AT2 3.py > ...
1  import pandas as pd
2  students = pd.read_csv("student.csv")
3  enroll = pd.read_csv("enrollment.csv")
4  merged = pd.merge(enroll, students, on="Student_ID")
5  print("Merged Data:\n", merged[['Student_Name', 'Department', 'Course_Name', 'Credits']])
6  credits = merged.groupby("Student_Name")['Credits'].sum()
7  print("\nTotal Credits per Student:\n", credits)

```

Output:

```

Merged Data:
  Student_Name Department Course_Name Credits
0    Raj Kumar        CS  Data Structures      4
1    Raj Kumar        CS   Algorithms      3
2    Priya S          IT Database Systems      4
3    Arun V           EEE    Circuits      3
4    Kavya R          MECH Thermodynamics      4
5    Kavya R          MECH  Fluid Mechanics      3

Total Credits per Student:
  Student_Name
Arun V        3
Kavya R        7
Priya S        4
Raj Kumar      7
Name: Credits, dtype: int64

```

4. Sales–Region Integration

You are given:

- sales.csv — Sale_ID, Product_ID, Region_ID, Quantity, Amount
- regions.csv — Region_ID, Region_Name, Manager

Task: Integrate the datasets using Region_ID. Display the region name, manager, product ID, quantity, and amount. Calculate the total sales amount for each region.

```

Sale_ID,Product_ID,Region_ID,Quantity,Amount
S001,P001,R01,10,5000
S002,P002,R02,5,3000
S003,P003,R01,8,4000
S004,P004,R03,12,6000
S005,P005,R02,7,3500

```

```

Region_ID,Region_Name,Manager
R01,North,John
R02,South,Priya
R03,East,Ravi

```

```

CO3 AT2 4.py > ...
1  import pandas as pd
2  sales = pd.read_csv("sales.csv")
3  regions = pd.read_csv("regions.csv")
4  merged = pd.merge(sales, regions, on="Region_ID")
5  print("Merged Data:\n", merged[['Region_Name', 'Manager', 'Product_ID', 'Quantity', 'Amount']])
6  region_sales = merged.groupby("Region_Name")['Amount'].sum()
7  print("\nTotal Sales per Region:\n", region_sales)

```

Output:

Merged Data:

	Student_Name	Department	Course_Name	Credits
0	Raj Kumar	CS	Data Structures	4
1	Raj Kumar	CS	Algorithms	3
2	Priya S	IT	Database Systems	4
3	Arun V	EEE	Circuits	3
4	Kavya R	MECH	Thermodynamics	4
5	Kavya R	MECH	Fluid Mechanics	3

Total Credits per Student:

Student_Name	
Arun V	3
Kavya R	7
Priya S	4
Raj Kumar	7

Name: Credits, dtype: int64

5. Weather–City Integration

You are given:

- cities.csv — City_ID, City_Name, State
- weather.csv — City_ID, Date, Temperature, Humidity, Rainfall

Task: Merge the datasets using City_ID. Display city name, state, date, temperature, humidity, and rainfall. Identify the city with the highest average temperature.

```
City_ID,Date,Temperature,Humidity,Rainfall
C001,2025-01-10,32,70,5
C002,2025-01-10,30,65,0
C003,2025-01-10,28,60,2
C004,2025-01-10,35,75,10
C001,2025-01-11,33,72,0
C002,2025-01-11,31,66,1
C003,2025-01-11,29,61,0
C004,2025-01-11,36,76,8
```

```
City_ID,City_Name,State
C001,Chennai,TN
C002,Coimbatore,TN
C003,Salem,TN
C004,Madurai,TN
```

Code:

```

CO3 AT2 5.py > ...
1 import pandas as pd
2 cities = pd.read_csv("cities.csv")
3 weather = pd.read_csv("weather.csv")
4 merged = pd.merge(weather, cities, on="City_ID")
5 print("Merged Data:\n", merged[['City_Name', 'State', 'Date', 'Temperature', 'Humidity', 'Rainfall']])
6 avg_temp = merged.groupby("City_Name")['Temperature'].mean()
7 highest_city = avg_temp.idxmax()
8 print("\nAverage Temperature per City:\n", avg_temp)
9 print("\nCity with Highest Average Temperature:", highest_city, "-", avg_temp.max())

```

Output:

```

Merged Data:
   City_Name State      Date  Temperature  Humidity  Rainfall
0   Chennai   TN  2025-01-10           32         70         5
1  Coimbatore   TN  2025-01-10           30         65         0
2    Salem   TN  2025-01-10           28         60         2
3   Madurai   TN  2025-01-10           35         75        10
4   Chennai   TN  2025-01-11           33         72         0
5  Coimbatore   TN  2025-01-11           31         66         1
6    Salem   TN  2025-01-11           29         61         0
7   Madurai   TN  2025-01-11           36         76         8

```

Average Temperature per City:

```

City_Name
Chennai      32.5
Coimbatore   30.5
Madurai      35.5
Salem        28.5
Name: Temperature, dtype: float64

```

City with Highest Average Temperature: Madurai - 35.5